Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		increases			1	1		
		(ii)		outer shell gets further away (from the nucleus) / there are more shells (1) there is lower attraction for an (incoming) electron / it is harder to gain an electron (1)	2			2		
	(b)			Cl_2 + 2NaBr → Br_2 + 2NaCl award (1) each for formulae Cl_2 NaCl award (1) for balancing only if both formulae are correct		3		3		
	(c)	(i)		3.20		1		1	1	
		(ii)		$\frac{1.27}{63.5}$ $\frac{3.20}{80}$ (1) 0.02 : 0.04 ⇨ 1: 2 ⇨ $CuBr_2$ (1) working **must** be shown ecf possible alternative method percentage composition of **both** elements		2		2	2	2

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				$\frac{1.27}{4.47} \times 100 = 28\%$ $\frac{3.20}{4.47} \times 100 = 72\%$ (1) for 100 g of compound $\frac{28}{63.5}$ $\frac{72}{80}$ 0.44 : 0.90 ⇨ 1:2 ⇨ $CuBr_2$ (1) ecf possible						
				Question 8 total	**2**	**6**	**1**	**9**	**3**	**2**